PROJECT SUMMARY · HINGLISH

Future Funds

"Paisa kahan ja raha hai?" se "kya main apne sapne tak pahunch paunga?" — ek living, goal-based financial roadmap.

India-focused personal finance planner · Educational guidance, regulated investment advice nahi · October 2026

1. Yeh project kya hai

Future Funds ek aisa app hai jo aapki poori financial life ko ek plan mein jod deta hai. Aap apni income, kharcha, EMIs, savings aur sapne daalte ho. App turant bata deta hai ki har goal ke liye kitna SIP chahiye, aaj kitna fund ho pa raha hai, aur mahine ka kitna gap bach raha hai.

Yeh bank account se link nahi hota. Koi fund ya stock recommend bhi nahi karta. Iska kaam decision clear karna hai: pehle kaunsa goal, kitna time, aur agar paise kam pad rahe hon to kya change karna chahiye. Isliye har screen ke neeche likha rehta hai ki yeh educational guidance hai, regulated investment advice nahi.

2. Kaise kaam karta hai

Socho ek salaried person, India mein, kai sapne ek saath: emergency fund, shaadi, ghar, bacche ki padhai, retirement. App unhe alag-alag calculators mein nahi todta. Ek hi plan banata hai.

1. **Profile.** Naam, age, retirement age, monthly income, doosri income, category-wise kharcha (rent, food, transport, bills, education, lifestyle), EMIs, cash, investments, gold, real estate, aur liabilities.
2. **Goals.** Template se goal add hota hai — emergency, retirement, education, house, wedding, business, car, travel, ya custom. Har goal ka aaj ka cost, target year, apni inflation, already saved amount, aur necessity (1 se 5) hota hai.
3. **Future cost.** Formula seedha hai: aaj ka cost × $(1 + \text{inflation})^{\text{years}}$. Education ki default inflation 10% hai, kyunki padhai general inflation se tez mehngi hoti hai. Ghar ke down payment par alag rate, emergency par alag.
4. **SIP.** Jo corpus chahiye, usme se pehle se save ki hui rakh ka future value nikal ke, baaki ke liye monthly SIP nikalta hai. SIP month ke start mein maana jaata hai, aur optional yearly step-up bhi hai — salary badhe to contribution bhi badhe.
5. **Surplus aur gap.** Investable surplus = income − kharcha − EMIs. Saare goals ka SIP jod ke, jo surplus se zyada ho, wahi monthly gap hai. Gap zero ya negative hai to plan funded hai.
6. **Priority.** Jab paise sabke liye kaafi na hon, rank banta hai: urgency × necessity × feasibility. Essentials (jaise emergency aur retirement) ko aage rakhne ka option hai. Surplus upar se neeche goals ko milta hai.
7. **Fix.** Gap page ek-click wale patches deta hai: kisi category ka kharcha kaato, SIP ko har saal 10% step-up karo, kam-priority goal ko peeche dhaka do, ya uska target chhota karo. Patch lagte hi poora plan dubara calculate hota hai.

> **Health score, 0 se 100.** Chaar hisson ka weight hai: savings rate (30 points, 30% savings par full), emergency fund (25 points, 6 mahine ke kharche par full), debt load (20 points, EMI income ke 40% se upar jaaye to score girta hai), aur goal coverage (25 points). Jis hisse ka data nahi hai, usse score mein nahi ginte — isliye khaali profile par health "—" dikhta hai, zero nahi.

3. App ke andar kya-kya hai

Sidebar chaar groups mein hai. Har page wahi ek plan padhta hai, alag data nahi.

Group	Page	Kya dikhata hai
Overview	Dashboard	Net worth, surplus, savings rate, health, insights, aur “improve health” actions.
Plan	Financial profile	Income, spending, assets, loans.
Plan	Goals	Inflation-adjusted target aur har goal ka SIP. Done goals alag record mein freeze ho jaate hain.
Plan	Life timeline	Age ke hisaab se milestones ek road par.
Analyse	What-if simulator	Income, expense, inflation, return badlo aur baseline se compare karo. Scenarios save ho sakte hain.
Analyse	Gap and fixes	Plan kahan short hai, aur one-click se kaise close ho.
Analyse	Goal prioritizer	Urgency, necessity, feasibility ke weights.
Analyse	Investments	Goal ke time horizon se glide path: equity, debt, gold, FD.
Analyse	Compounding lab	Delay ki cost, catch-up SIP, rule of 72.
Analyse	Risk lab	Monte Carlo. Har goal ki success probability, p10 / p50 / p90 fan chart ke saath.
Assist	Financial news	India aur global, aaj aur kal. Read / unread isi browser mein rehta hai.
Assist	AI advisor	Plan ke baare mein sawal. Numbers model khud nahi nikalta.
Assist	Settings and data	Assumptions, JSON export / import, print, reset.

4. Architecture — hisaab kahan hota hai

Sabse important design decision yeh hai: **saara maths browser mein chalta hai.** Slider hilate hi plan dubara banta hai, server ka wait nahi. Wahi formulas Python mein bhi likhe hain, taaki AI advisor jab tool call kare to number guess na kare.

Browser, hamesha

React UI state padhta hai, src/engine plan banata hai, Zustand us state ko browser mein save karta hai. Dashboard se lekar Monte Carlo tak, yahi plan use hota hai.

Backend, optional

FastAPI sirf tab chahiye jab AI chat ya financial news chahiye. Bina key ke bhi app khulta hai. Advisor ke paas local fallback hai, jo seedha plan se jawab likh deta hai.

Data aapka, machine par

Profile, goals, assumptions, scenarios, chat, aur news read-marks **localStorage** mein rehte hain, key **future-funds-v1**. Koi login nahi. JSON export / import aur print-to-PDF settings mein hai.

Do engines, ek matlab

TypeScript engine UI ke liye hai. Python engine advisor ke tools ke liye hai: future cost, SIP, probability, delay cost. Dono ka contract same hai, isliye model ko calculate karne ki ijaazat nahi.

Request ka rasta: browser **/api** call karta hai. Vite usko backend tak proxy kar deta hai (local par port 8000, Docker mein service name **backend**). CORS default sirf **http://localhost:5173** ko allow karta hai.

5. Technologies aur components

Layer	Technology	Kaam
UI	React 18	Pages, charts, forms. Single-page app, route library nahi — page Zustand mein switch hota hai.
Language	TypeScript 5	Profile, goal, plan, patch, allocation — saare types ek jagah.
Build	Vite 5	Dev server port 5173, React plugin, **/api** proxy.
State	Zustand 4 + persist	Profile, goals, completed goals, assumptions, scenarios, chat, news reads. Auto-save localStorage.
Charts	Recharts 2	Cashflow, SIP bars, allocation, projection, Monte Carlo fan.
API	FastAPI + Uvicorn	Health, news, goal calc, Monte Carlo, advisor chat.
Validation	Pydantic 2	Request bodies: chat, goal calc, Monte Carlo.
Config	python-dotenv	**AI_PROVIDER**, API keys, Marketaux token, CORS.
AI	Gemini (default) ya Claude	**google-genai** ya **anthropic**. Provider **.env** se switch hota hai.
News	Marketaux	India aur global headlines, aaj + kal. Free quota bachane ke liye server par 1 hour cache.
Tests	pytest	Engine, advisor, aur news ke tests backend mein.
Containers	Docker Compose	Frontend: Node 20 Alpine. Backend: Python 3.12 slim. Ports 5173 aur 8000.

Code ke main hisse

- **frontend/src/engine** — finance (future cost, SIP, lump sum), plan (surplus, rank, gap, fixes), health, health actions, insights, patch, Monte Carlo.
- **frontend/src/pages** — upar wali saari screens. Layout sidebar se page choose karta hai.
- **frontend/src/data/templates.ts** — goal templates, default assumptions (inflation 6%, equity 12%, debt 7%, FD 6.5%, gold 8%), aur sample profile.
- **frontend/src/store/useStore.ts** — saari user state aur JSON import.
- **backend/app/engine.py** — wahi maths, Python mein, advisor ke liye.
- **backend/app/advisor.py** — system prompt, tools, Gemini/Claude call, aur key na ho to plan-only jawab.
- **backend/app/news.py** — Marketaux fetch, India timezone windows, hourly cache.

6. AI advisor aur news, thoda kareeb se

Advisor ko instruction hai: pehle **get_plan_summary** lo, phir kisi bhi figure ke liye tool use karo. Woh khud multiply nahi karta. Jawab short hota hai, rupee lakh/crore format mein, aur concrete options deta hai — SIP badhao, kharcha kaato, deadline badlo, target kam karo. Specific mutual fund ya stock ka naam nahi deta.

Key missing ho, ya provider set galat ho, to chat phir bhi chalta hai. Backend plan summary se hi gap, priority, aur emergency fund ka jawab bana deta hai, aur saath mein likh deta hai ki yeh finance engine se aaya hai.

News Marketaux se aata hai, sirf aaj aur kal, India scope aur global scope. Browser apna timezone offset bhejta hai (India ke liye default IST). Padha hua article isi browser mein mark rehta hai, aur purane read-marks ek hafta baad saaf ho jaate hain. Token na ho to news page clearly bata deta hai ki feed configured nahi hai.

> **Default assumptions, jo Settings se badal sakte ho.** General inflation 6%. Equity return 12% (volatility 18%), debt 7% (4%), gold 8% (14%), FD 6.5%. SIP step-up default 0%. Monte Carlo default 300 runs. SIP ko thodi conservative return par size kiya jaata hai, taaki fully funded goal ki success chance kareeb 80% ke aas-paas rahe. Returns assumption hain, guarantee nahi.

7. Chalane ka tareeka

Backend optional hai. AI chat aur news ke liye chahiye.

- Backend: **backend** folder mein venv, **pip install -r requirements.txt**, **.env** mein **AI_PROVIDER=gemini** aur **GEMINI_API_KEY**. Claude chahiye to **AI_PROVIDER=anthropic**. News ke liye **MARKETAUX_API_TOKEN**. Phir **uvicorn app.main:app --reload --port 8000**. Tests: **pytest**.
- Frontend: **frontend** folder mein **npm install**, phir **npm run dev**. App **http://localhost:5173** par khulta hai.
- Dono ek saath: project root se **docker compose up --build**.

8. Short mein yaad rakhne layak baat

Future Funds ek calculator ka collection nahi hai. Yeh ek plan hai jo income, kharcha, aur kai goals ko ek surplus par ladta hua dikhata hai. Jo number screen par hai, woh browser engine ka hai. Jo number advisor bolta hai, woh bhi usi engine ka hai — model sirf us number ko aam zabaan mein samjhata hai. Data aapke browser mein rehta hai. Aur poora product India ke SIP, EMI, lakh-crore, aur education-inflation wale context ke liye likha gaya hai.